PID-Based Anomaly Detection in Federated Learning

Experimental Evaluation Under Label-Flipping Attacks

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Abstract— Twenty FEMNIST clients were evaluated under a Group 5 label-flipping attack with ten clients designated poisoned. The presented Proportional-Integral-Derivative (PID) defense monitors client model updates for anomalies and is intended to exclude suspicious contributors. In the supplied ten-round results, however, no clients were removed. The final plotted average accuracy was approximately 29.0% for conventional federated learning (FL) versus 20.5% for PID-enhanced FL; loss was approximately 2.07 versus 2.286. The observed configuration therefore did not demonstrate successful quarantine or improved attacked performance. We analyze the trends, inspect partial Flower logs, and identify experiments required to determine why exclusion did not occur.

Index Terms— Federated learning, PID controller, data poisoning, label flipping, anomaly detection, FedAvg.

I. INTRODUCTION

Federated learning (FL) aggregates model updates trained locally without directly centralizing client data. Its aggregation stage remains vulnerable when adversaries manipulate labels or updates. This study evaluates a PID-based anomaly detector against conventional Federated Averaging (FedAvg), using group-specific poisoning conditions and round-wise performance metrics. The evaluation focuses on accuracy, loss, per-client behavior, attack severity, and exclusion dynamics.

II. EXPERIMENTAL DESIGN

The documented implementation uses the instructor-provided FEMNIST digit subset distributed over 20 virtual clients with an 80/20 local train/test split (seed 42). A DigitCNN has convolutional layers with 32 and 64 filters, max pooling, a 128-unit dense layer, dropout 0.2, and ten output classes. The prior implementation describes Flower/Ray simulation, Adam (learning rate 0.001), batch size 32, and one local epoch per round. The supplied plots cover ten rounds. These configuration details are inherited from the prior implementation description rather than verified against current source code.

The Group 5 threat model designates 50% of clients as poisoned (ten of twenty). The documented attack transforms each poisoned target label as $y_{attack} = (y + 1) \bmod 10$, with 100% label flipping for those clients. The attack-severity experiment additionally plots four and sixteen poisoned clients, but matching run settings and replicates were not provided.

III. PID DETECTOR AND QUARANTINE RULE

The detector forms a centroid of active client weight vectors and computes each client's normalized Euclidean distance $D_k(t)$ from the centroid. For client k at round t, its temporal anomaly score is:

$$\mathbf{u}_k(t) = K_p D_k(t) + K_i \Sigma_{i=1..t} D_k(i) + K_d[D_k(t) - D_k(t-1)].$$

The first round uses only the proportional term because historical terms are unavailable. Earlier documentation reports $K_p = 1.0$, $K_i = 0.05$, $K_d = 0.5$, and a threshold $\tau = 0.45$. A client exceeding or meeting the threshold should be removed from aggregation and quarantined. Current source code and individual PID score traces were not supplied; the exact active settings and reason for zero exclusions therefore remain unverified.

IV. COMPARATIVE PERFORMANCE

Table I summarizes round-ten values approximated from the graph markers. These values are plot estimates, not exact exported CSV measurements.

TABLE I. ROUND-TEN MEASURE	CONVENTIONAL	PID
Average accuracy	~0.290 (29.0%)	~0.205 (20.5%)
Average loss	~2.07	~2.286
Poisoned clients excluded	Not reported	0 of 10

Conventional FL finished roughly 8.5 percentage points higher in accuracy and about 0.22 lower in loss. Its accuracy rises unevenly across the experiment. PID accuracy stays near 10.4% during rounds 2–8, then increases to approximately 15.3% and 20.5% in rounds 9 and 10. Thus the plotted defense does not exceed conventional FedAvg on either reported final metric.

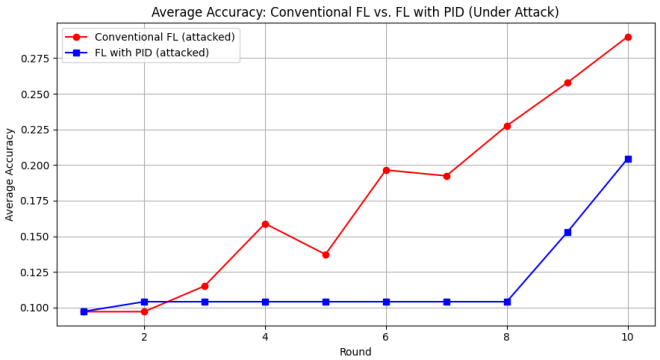


Fig. 1. Average accuracy across ten attacked rounds: FedAvg versus PID-enhanced FL.

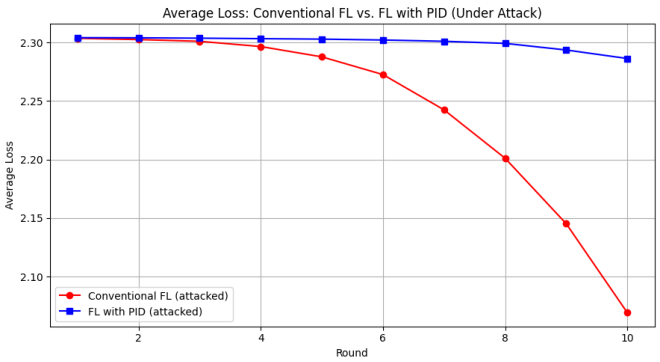


Fig. 2. Average evaluation loss across ten attacked rounds.

V. CLIENT-LEVEL LEARNING TRAJECTORIES

The client-specific PID accuracy traces differ appreciably across clients. Several series remain largely flat from rounds 2 to 8 and increase only near the end, suggesting delayed or limited learning rather than early convergence. Mean accuracy is approximately 0.104 throughout that plateau and reaches about 0.205 at round 10.

Individual loss trajectories cluster around 2.29–2.31 in early rounds, with slight decreases later. The average loss declines from around 2.304 to about 2.286. Both the individual trajectories and the mean show relatively small loss improvements compared with the conventional attacked run.

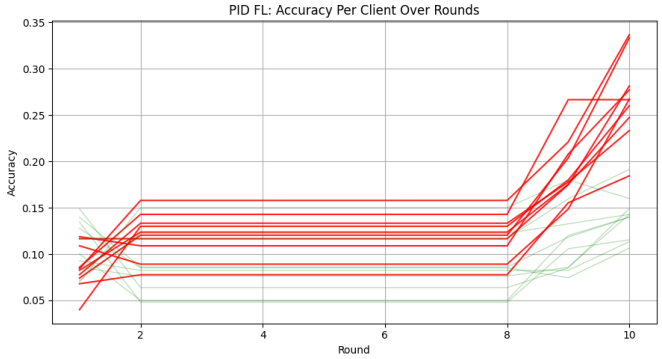


Fig. 3. Accuracy of individual clients in the PID experiment.

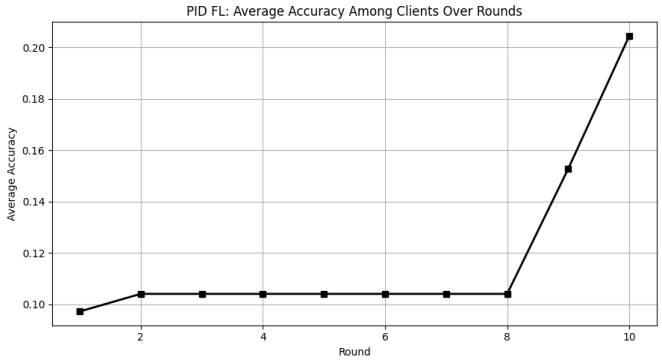


Fig. 4. Mean PID client accuracy over rounds.

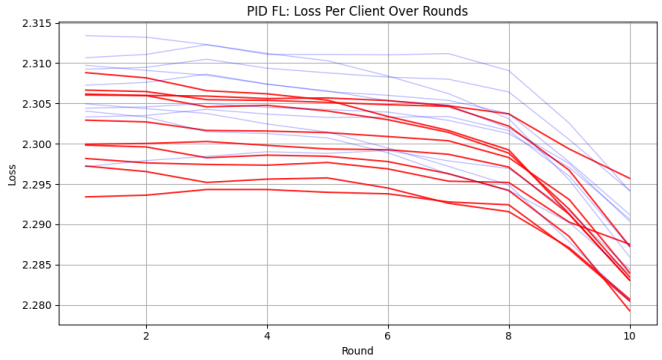


Fig. 5. Loss trajectories for individual PID clients.

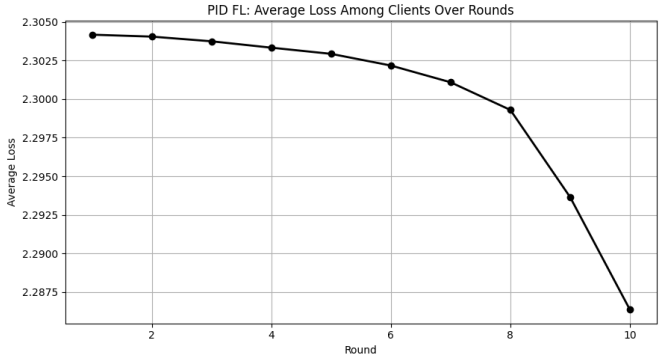


Fig. 6. Mean client evaluation loss in the PID experiment.

VI. ATTACK SEVERITY AND CLIENT EXCLUSION

At four, ten, and sixteen poisoned clients, the reported final PID accuracies are approximately 35.6%, 10.6%, and 27.7%, respectively (Fig. 7). The pattern is non-monotonic; the figures alone cannot identify whether random initialization, client mix, sample composition, or differing run settings explain this outcome. No error bars or repeated-run statistics were supplied.

Figure 8 records zero cumulative removals in every round, versus ten ground-truth poisoned clients. Assuming the graph refers to the same ten-poisoned-client experiment, detected poisoned-client recall is 0/10 = 0%. This observation contradicts the historical claim of all malicious clients being quarantined by round 2. The current evidence does not show whether PID scoring, threshold calibration, exclusion execution, or plotting instrumentation produced the failure.

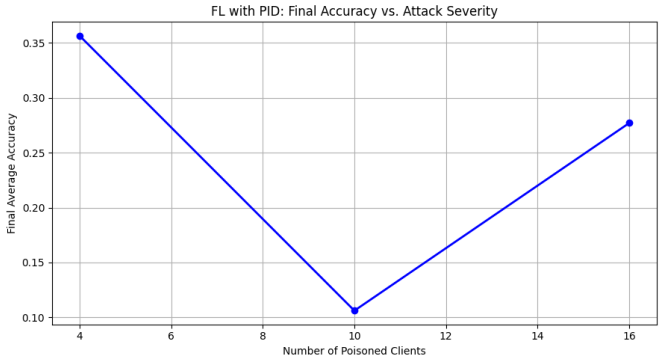


Fig. 7. PID final accuracy versus poisoned-client count.

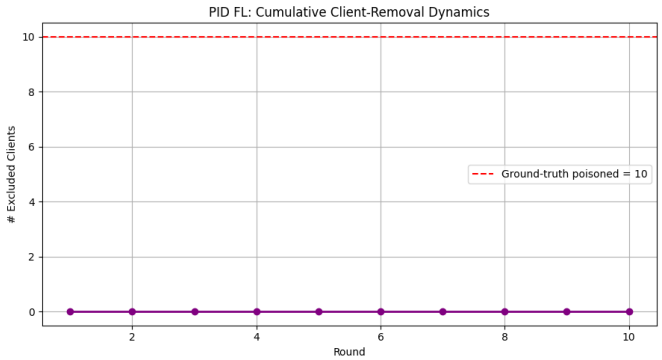


Fig. 8. Cumulative excluded clients versus the ten poisoned clients.

VII. FLOWER CONSOLE EVIDENCE

The partial Flower transcript contains two five-round simulation summaries. During rounds 1–5 in both excerpts, the strategy reports “20/20 clients kept; newly excluded=[]” and each aggregate-fit step receives 20 results with zero failures. One pass completes in 39.21 s and the other in 38.62 s. The two summaries are not necessarily consecutive halves of one uninterrupted ten-round training history.

ROUND	RUN A LOSS	RUN B LOSS
1	2.3039497267	2.3035474058
2	2.3036857294	2.3019168046
3	2.3032067047	2.2965008948
4	2.3029994764	2.2848019184
5	2.3004344642	2.2630602700

The excerpt also shows missing custom fit/evaluation metric aggregation callbacks and deprecation warnings concerning the legacy client_fn(cid) signature and start_simulation() API. These

do not prove a detector defect, but raise maintenance and measurement-consistency issues. It is not established that the console excerpt exactly matches all charted runs.

VIII. DISCUSSION AND LIMITATIONS

The available evidence supports a narrow conclusion: the plotted PID run excluded no poisoned clients and underperformed conventional FL in final attacked accuracy and loss. Earlier claims of 75.4% PID accuracy, 0.982 loss, and 100% malicious-client exclusion cannot be attributed to the supplied results. They may concern another experiment, but no raw artifacts establish correspondence.

The detector should be instrumented to export per-client weight distances, proportional/integral/derivative score contributions, final PID scores, thresholds, decisions, and ground-truth poisoning status. Aggregation should be checked to ensure rejected updates cannot enter FedAvg and quarantined clients cannot reappear. Each figure, CSV, and log should share a run identifier, random seed, threat-model parameters, and metric definition.

Controlled reruns should compare clean and attacked conditions using matched partitions and multiple seeds; report mean and variance, precision, recall, and false positives; and evaluate threshold and gain sensitivity. Per-client averages should be distinguished from centrally evaluated global-model metrics. Missing clean-run curves, score traces, CSV data, and uncertainty intervals are not inferred here.

IX. IMPLEMENTATION AND CONTRIBUTIONS

The codebase includes src/dataset.py (FEMNIST data and label corruption), src/model.py (DigitCNN), src/pid_strategy.py (PID scoring and exclusion), and src/train_pid.py (simulation, logging, and plots). From the project root, run: python src/train_pid.py. The script saves PID results to results/pid_attack and comparison plots to results/comparison.

CONTRIBUTOR	DOCUMENTED RESPONSIBILITY	HOURS
Tyriz Newton	PID detector, weight distances, scoring, drafting	9
Kai Fan	Removal tracking, metrics, plots, analysis	9
Both	Simulation, tuning, review	16 (joint)

The contribution assignments and hours are carried forward from the previous paper and should be confirmed against current records.

X. CONCLUSION

The ten-round experiment does not demonstrate effective malicious-client exclusion by the evaluated PID configuration. No poisoned clients were removed, and PID-enhanced FL ended with lower accuracy and higher loss than conventional FL under attack. A transparent negative result provides a basis for instrumenting the detector and performing a controlled, reproducible re-evaluation.

REFERENCES

[1] Rochester Institute of Technology, Federated Learning Privacy and Security Enhancement, CSCI 532 course project specification.

[2] T. Newton and K. Fan, Enhanced Federated Learning via PID-Based Anomaly Detection and Malicious Client Exclusion, previous Part 3 paper.

[3] Experimental plots and client-removal diagrams, eight supplied PNG figures.

[4] Partial Flower/Ray execution transcript, supplied console output.

[5] H. B. McMahan et al., “Communication-efficient learning of deep networks from decentralized data,” AISTATS, 2017.

[6] D. J. Beutel et al., “Flower: A friendly federated learning research framework,” arXiv:2007.14390, 2020.